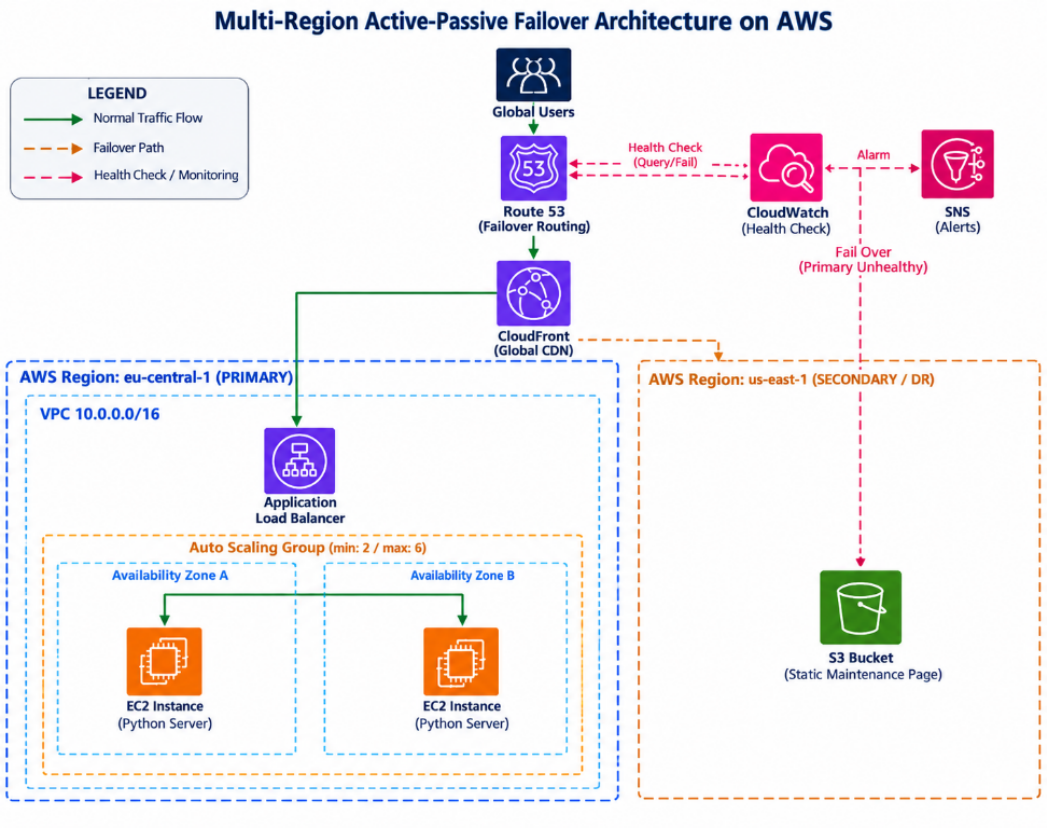
Task 1.1.2 | Development and reflection

- Multi-region active-passive failover webpage on AWS.
- This presentation distinguishes the live static explainer from the separate Terraform prototype.
- The scenario controls illustrate request paths in the browser; they do not change AWS.
- Current evidence and limitations are stated explicitly rather than inferred from design diagrams.

Problem, audience, and objectives

- Explain how a web request can reach an AWS-hosted site and what a regional-failure response could look like.
- Primary goals: clear visual communication, repeatable Terraform, HTTPS delivery, and scalable application capacity.
- The 99.99% availability target is a design aspiration, not a measured result or provider guarantee.
- The sample has no database, session replication, or durable user data.

Architecture at a glance



Conceptual diagram; it does not represent deployed resources.

- Normal path: visitor → CloudFront → primary ALB → EC2 Auto Scaling application.
- Origin failure: selected ALB 5xx responses can cause CloudFront to request the secondary S3 maintenance page.
- Route 53 DNS failover appears in the proposed concept but is not present in the Terraform prototype.

Live product and scope boundary

- The public CloudFront URL hosts the static interactive explainer.
- The live static-site deployment uses a private S3 origin and CloudFront Origin Access Control.
- The live website is not the multi-region application and does not prove application failover.
- The separate multi-region Terraform root has not been applied in AWS.

Implementation: primary region

- A VPC contains two public subnets in distinct available Availability Zones.
- An internet-facing ALB forwards to a target group using the /health endpoint.
- The launch template installs a small threaded Python HTTP service managed by systemd.
- The application runs as an unprivileged user; instance metadata requires IMDSv2.

Implementation: scaling and failover

- Auto Scaling defaults: minimum 2, desired 2, maximum 6, CPU target 50%.
- CloudFront origin-group failover is configured for HTTP 500, 502, 503, and 504.
- The secondary S3 website provides a maintenance response; it is directly reachable over HTTP.
- A CloudWatch unhealthy-target alarm publishes to an SNS topic; email is optional and requires confirmation.

Security and operating considerations

- The ALB accepts origin traffic from CloudFront's managed origin-facing prefix list.
- Application instances accept HTTP only from the ALB security group.
- The secondary website intentionally permits public reads of index.html only; never store sensitive content there.
- The multi-region stack creates billable resources and requires an approved AWS identity, plan review, and cost check.

Development and validation

- The project history records Terraform 1.12.2 formatting and provider-schema validation as passing.
- The history also records eight mocked, plan-only test runs; they create no cloud resources.
- Checks cover selected inputs, capacity, network boundaries, health checks, IMDSv2, origin responses, and alarm wiring.
- Terraform was unavailable during this package rebuild, so these recorded checks were not rerun.

Reflection: limitations and DNS design

- The multi-region stack has no current AWS plan, apply, runtime-failover, alarm-delivery, or recovery evidence.
- Route 53 records, DNS health checks, a controlled domain, and custom-domain certificates are not implemented.
- Direct DNS failover to an S3 website endpoint would not preserve HTTPS.
- A future design must test health-check reachability, HTTPS on both targets, DNS caching, recovery, and rollback.

Finalisation and next steps

- The PDF and ZIP package is rebuilt from current editable sources; historical deployment claims are not reused as evidence.
- Next: obtain approval and credentials, review cost and plan, deploy in an isolated account, and collect timestamped test results.
- Then test origin failure, alarms, restoration, and measured recovery; only afterward complete and test the DNS layer.
- Project: https://github.com/EunatiaN1/multi-region-active-passive-failover
- Live static explainer: https://d2quyljtx3dqoe.cloudfront.net/